Graph Based Recommendations on MovieLens 100K

Personalized PageRank, Retrieval Accuracy, and MF Continuity

Student: Abaidullah Seikh Team: Individual

Date: 2026-10-29 Assignment: A06 Graph Based Recommendations (Week 6)

Project: https://abaidullahseikh.github.io/RecSys-LLMs/week%206/
Repository: https://github.com/abaidullahseikh/RecSys-LLMs/tree/main/week%206

Abstract. I implemented the seven required graph and recommendation functions and compared popularity, global PageRank, Personalized PageRank (PPR), and the supplied HW6 matrix factorization (MF) reference on MovieLens 100K. All four methods used 861 shared test users and 1,611 training movie candidates. Validation selected PPR α = 0.50; its test Recall@10 was 0.11219, compared with PageRank 0.07969, popularity 0.07602, and browser MF 0.02114. The MF reference selected epoch 1 by validation RMSE but produced identical candidate scores in this run, so it is not a trustworthy substitute for the unavailable original Week 5 model. Two users and controlled restart changes illustrate list differences, not satisfaction gains. I recommend testing PPR in place of this reference MF for the main retrieval section before deployment.

Index Terms: recommender systems, MovieLens 100K, PageRank, personalized random walks, Recall@10, matrix factorization

1. Introduction

Problem statement. The assignment requires exactly seven functions: buildGraph, rowNormalize, pageRank, personalizedPageRank, movieScores, recommendTopK, and recallAtK. They must work with the supplied browser application and TensorFlow.js. The business task is to decide whether a streaming platform should keep MF, replace it with PPR, or use both [7].

Motivation. A popular movie is not automatically the right movie for every person. Global PageRank emphasizes overall connectedness, whereas PPR can restart a walk at a specific user. MF instead learns latent preferences from ratings. These methods answer related but different questions, so a fair decision needs a shared candidate pool, a shared user cohort, correct filtering, and a measured Top 10 metric rather than a comparison of unlike score scales [4,7].

Contributions. I implemented and verified the seven HW6 functions, tested dangling mass and the four node reference graph, selected α using validation Recall@10, and audited all four methods against the browser exports. The supplied MF reference was evaluated as recorded; its constant scores and lack of continuity with the original Week 5 checkpoint are disclosed, not concealed.

2. Related Work

MovieLens provides explicit ratings, but its timestamps do not describe viewing sessions [1]. PageRank models graph importance by propagating probability across links [2], and topic-sensitive PageRank motivates biased restart distributions [3]. Matrix factorization represents users and movies with learned preference vectors [4]. Rating accuracy and ranked retrieval answer different questions, which is why I report both rather than using either as a substitute for the other [7]. TensorFlow.js provides the browser tensor operations used in this study [5].

3. Method

3.1. Approach and Pipeline

I loaded the original MovieLens files, applied the chronological split per user, built a graph from positive training ratings, normalized outgoing transitions, and ran PageRank and PPR. Validation selected the link probability before test evaluation. Four methods used the same candidate pool and shared users: popularity counts distinct positive training raters (rating at least four); global PageRank uses uniform restart; PPR restarts at the mapped target user; and the supplied HW6 MF reference ranks training mean plus the user and movie latent factor dot product. PPR movie node scores are stationary walk probabilities, neither ratings nor purchase probabilities, and cannot be compared in scale with MF scores. I verified fixtures, common cohort membership, exported recommendations, and controlled restart changes [7].

3.2. Data Preparation, Graph and Transitions

MovieLens 100K has 100,000 ratings from 943 users on 1,682 films; the observed rating matrix is 93.6953% sparse [1]. The chronological split within each user contains 79,619 training, 9,596 validation, and 10,785 test records, with no records removed. It is not a global chronological split, and missing ratings are unobserved. Training includes 943 user IDs and 1,611 movie IDs, mapped in ascending order to 2,554 contiguous graph nodes without padding. Exactly 45,602 distinct training ratings of at least four create undirected user to movie edges. User 685 and 253 training movies have degree zero.

For training user u and movie i, R[u,i] = 1 exactly when the observed training rating is at least 4; otherwise R[u,i] = 0. Zero means no positive interaction, not dislike. The adjacency is A = [[0,R],[R^T,0]]: both user to movie and movie to user movement are allowed, but no user to user or movie to movie edges exist. Nonzero adjacency rows are divided by degree to form link following probabilities. If the transition matrix P has no dangling rows, the update is p(next) = α P^T p + (1 − α)v. For isolated nodes, the implementation uses:

$$p(\text{next}) = \alpha\,(Q^T p + d\,v) + (1-\alpha)v, \qquad p(0) = v.$$

Here α is the probability of following a link, 1 − α is the restart probability, and v is a nonnegative restart distribution summing to one. Q keeps normalized rows for positive degree nodes and zero rows elsewhere; d is the current probability mass at zero degree nodes. The transpose accumulates incoming mass without reversing the graph direction. Returning d through v conserves total probability in both PageRank and PPR. Global PageRank has a uniform v; PPR uses a one hot v at the chosen user node. Starting from p = v, iterations stop at an L1 change below 10^{-6} or after 500 updates. Hitting the cap alone does not mean convergence [2,3,7].

3.3. TensorFlow.js Implementation and Design Decisions

The browser uses TensorFlow.js 4.22.0 and vendored D3 7.9.0. After tf.ready(), tf.getBackend() showed WASM for graph runs and WebGL for MF training. Probability propagation uses tf.sparse.sparseSegmentSum on WASM or rectangular R and R^T blocks with tf.matMul elsewhere, without a full square adjacency. tf.tidy() manages temporary tensors; long lived tensors are disposed and async callbacks are not placed inside tidy. The saved browser run took 47.60 s overall, including 39.22 s for the four method comparison; tensor counts remained at 10. Times are for this recorded environment only [5].

Recommendation protocol. I score only the 1,611 training movie candidates, extracting movie nodes for PageRank and PPR. For TEST, I exclude every movie the target user rated in training or validation, regardless of rating; for VALIDATION, I exclude training rated movies only. Remaining movies are sorted by decreasing score, with ties resolved by ascending movie ID. Each list includes up to ten IDs, titles, genres, scores and positive training popularity. All audited primary test lists contained ten movies and none needed a zero score entry. A zero score included through tie breaking would not prove relevance. Test ratings remain held out and eligible to be retrieved [7].

3.4. Four Methods and Shared Evaluation

Recall@10 is the macro average of each eligible user's retrieved relevant films divided by all eligible relevant films: held out ratings of at least four, present in the training movie pool, and passing the history mask. The exported popularity, PageRank, PPR and browser MF rows all use the same 861 test users and 1,611 candidates with identical exclusion rules, so there is no MF cohort mismatch. Among 943 users, 82 lack eligible test relevant movies, including zero degree user 685; thirteen distinct relevant test movie IDs are outside the pool. Validation uses 846 eligible users; 97 are excluded and four relevant movie IDs are unavailable. This is warm user offline evaluation, not evidence of cold start performance.

The primary fourth method is the MF reference run by the submitted HW6 browser, not a separate Week 5 reproduction. It uses 16 factors, seeded initialization, Adam (rate 0.01), batch size 512, regularization 0.05 and up to 30 shuffled epochs. The validation RMSE was 1.182994 throughout; the earliest best checkpoint was epoch 1. The scorer is training mean plus user and item factor dot product, but every test candidate scored 3.580603 in the recorded run. Consequently, its Top 10 lists are sorted by eligible ascending movie ID. The original Week 5 selected weights were unavailable, and the supplied MF training objective differs; this comparison cannot establish performance against the true previous Week 5 model.

4. Experiments

4.1. Validation and Experimental Controls

I tested α = 0.50, 0.70, 0.85 and 0.95 using the same 846 validation users (Table 2). The selected α = 0.50 maximized Recall@10, with smaller α winning exact ties; test results were not used. Global PageRank uses α = 0.50 as a shared control, not an independent optimum. The supplied HW6 MF reference chose epoch 1 by validation RMSE, a different selection objective.

Table 1. Dataset, Model and Shared Evaluation Controls

Measure	Verified value	Measure	Verified value
Users / catalog movies	943 / 1,682	Graph nodes / candidates	2,554 / 1,611
Train / validation ratings	79,619 / 9,596	Test ratings	10,785
Positive training edges	45,602	Zero degree users / films	1 / 253
Validation cohort	846	Test cohort	861
Unseen relevant film IDs, val / test	4 / 13	Excluded users, val / test	97 / 82
Sparsity / relevance threshold	93.6953% / rating ≥ 4	Top 10 candidate source	Training movies

Table 2. PPR Validation Sweep on the Shared 846 User Cohort

Link probability α	Validation Recall@10	Convergence note
0.50	0.114483	Selected; converged
0.70	0.104797	Converged
0.85	0.090021	Converged
0.95	0.073750	27 users reached iteration limit

Table 3. Consolidated Test Comparison, Shared 861 Users

Method	α / epoch	Recall@10	Avg. training popularity
Popularity	—	0.076023	272.00
Global PageRank	α = 0.50	0.079693	257.82
Personalized PageRank	α = 0.50	0.112189	244.21
Supplied HW6 MF reference	epoch 1	0.021137	96.76

Result. PPR improves absolute Recall@10 by 0.03250 over global PageRank, 0.03617 over popularity, and 0.09105 over the supplied HW6 MF reference, on the same 861 users. These are measured differences in offline retrieval, not satisfaction or revenue gains. MF's low popularity follows score ties and ID ordering, not demonstrated discovery quality.

4.2. Verification: Sanity and Independent Checks

Before the MovieLens runs, I tested the directed four node reference graph with edges A to B, A to C, B to C, C to A, and D to C. A splits its outgoing mass equally between B and C, and none of the four nodes is dangling. At α = 0.85, uniform restart produced p = [0.3725268513, 0.1958239118, 0.3941492369, 0.0375000000], ranked C > A > B > D. Restart only at A produced p = [0.4522328999, 0.1921989825, 0.3555681176, 0.0000000000], ranked A > C > B > D. Both matched the specified reference values, which are fixture results rather than MovieLens results [7].

Table 4. Verification Evidence and Observed Outcome

Check	Observed result
Browser harness and integration	Final verification.json: 12 passed, 0 failed, 0 pending. Separate OpenCode integration log: 33 passed, 0 failed.
Normalized rows	6 positive-degree rows summed to 1; 2 dangling rows stayed zero
Restart distributions	Uniform and one-hot vectors were nonnegative and summed to 1
Probability and score sanity	3-node probability sum 1.0000002682; stationary scores nonnegative
One extra stationary update	L1 difference 7.972×10^{-7}, below the 10^{-6} threshold
Separate zero-degree fixture	Mask [false, true, false, false, true]; mass ≈ 1 preserved
Direction and candidate filtering	Movie nodes; TEST masks train and val, VALIDATION masks train; ties by ID; no short or zero score test lists
MovieLens convergence, α = 0.50	PageRank 19 steps, residual 5.37×10^{-7}; PPR 21–22 steps
Independent four-method ranking	Recall@10 reproduced; 861 identical test users and 1,611 movies
Browser MF reference checkpoint	Epoch 1; validation RMSE 1.182994; constant test candidate scores 3.580603
MF state during evaluation	Saved factor fingerprints unchanged; primary CSVs match browser method and cohort

At α = 0.95, 27 validation users reached the 500 iteration cap without convergence; the selected α = 0.50 had no such failures. For a hand-checkable retrieval example, user 1 has 19 eligible relevant test films and PPR retrieves three (IDs 100, 258 and 222), so Recall@10 = 3/19 = 0.15789. User 2 has two eligible relevant films and no hits, giving 0/2 = 0. The PPR advantage therefore does not apply to every user.

5. Discussion

5.1. Failure Case

A PPR integration check restarted at node 1 for user ID 1, but that user maps to node 0. Also, 27 validation walks at $\alpha = 0.95$ hit the 500 iteration cap without convergence. The supplied HW6 MF reference gave all candidates identical scores.

5.2. Root Cause

Raw user IDs are not zero based graph node indices, so directly using an ID seeds the wrong user. High α caused slow convergence for some walks. The supplied MF factors collapsed to indistinguishable scores; its ranking fell back to movie ID rather than learned preference order.

5.3. Fix and Verification

I fixed the PPR adapter with graph.nodeOfUser() and unknown user checks. User 1 then matched a direct node 0 restart exactly; the incorrect node 1 restart differed by 0.1529675. Four node vectors, mass conservation, dangling cases, matrix orientation and filtering were tested. verification.json reports 12 / 0 / 0 and a separate OpenCode suite 33 / 0. All selected $\alpha = 0.50$ test walks converged. The 27 capped $\alpha = 0.95$ walks and the unchanged degenerate MF reference are disclosed.

5.4. Two Users and Their Histories

For fixed test users 1 / 2, training had 217 / 49 ratings and 131 / 32 positives. Frequent positive genres were Drama (61 / 18), Comedy (39 / 11), plus Action (36) for user 1 and Romance (11) for user 2. Tables 5–6 give complete PPR Top 10 lists; root recommendations.csv gives full scored Top 10 lists for all four methods.

Table 5. User 1 PPR Top 10 (Selected $\alpha = 0.50$)

R	Film (movie ID)	Genre(s)	PPR score	Pop.
1	Fargo (100)	Crime/Drama/Thr.	0.000686	377
2	Contact (258)	Drama/Sci Fi	0.000440	307
3	English Patient, The (286)	Drama/Rom./War	0.000431	279
4	Schindler's List (318)	Drama/War	0.000404	224
5	One Flew Over the Cuckoo's Nest (357)	Drama	0.000377	202
6	Casablanca (483)	Drama/Rom./War	0.000360	197
7	Star Trek: First Contact (222)	Action/Adv./Sci Fi	0.000333	189
8	Trainspotting (475)	Drama	0.000328	164
9	Sense and Sensibility (275)	Drama/Rom.	0.000320	178
10	E.T. the Extra-Terrestrial (423)	Child./Drama/Fant./Sci Fi	0.000306	167

Table 6. User 2 PPR Top 10 (Selected $\alpha = 0.50$)

R	Film (movie ID)	Genre(s)	PPR score	Pop.
1	Return of the Jedi (181)	Action/Adv./Rom./Sci Fi/War	0.000697	339
2	Dead Man Walking (9)	Drama	0.000498	182
3	Twelve Monkeys (7)	Drama/Sci Fi	0.000474	234
4	Rock, The (117)	Action/Adv./Thr.	0.000456	215
5	Silence of the Lambs, The (98)	Drama/Thr.	0.000447	308
6	Independence Day (ID4) (121)	Action/Sci Fi/War	0.000431	207
7	Raiders of the Lost Ark (174)	Action/Adv.	0.000403	312
8	Chasing Amy (268)	Drama/Rom.	0.000402	146
9	Trainspotting (475)	Drama	0.000402	164
10	Mr. Holland's Opus (15)	Drama	0.000401	160

Rank 1 for users 1 / 2 (genre; score; positive training popularity): popularity: Fargo (Crime/Drama; 377; 377) / Return of the Jedi (Action/Sci Fi; 339; 339); PageRank: Contact (Drama/Sci Fi; 0.002375; 307) / Return of the Jedi (Action/Sci Fi; 0.001989; 339); MF: Copycat (Crime/Thriller; 3.580603; 28) / GoldenEye (Action/Thriller; 3.580603; 41). PPR ranked Fargo (0.000686; 377) / Return of the Jedi (0.000697; 339) first (Tables 5–6). Mean Top 10 popularity for users 1 / 2: popularity 246.8 / 262.1; PageRank 239.4 / 247.4; PPR 228.4 / 226.7; MF 72.2 / 111.4. Cross-user overlaps: 1, 1, 1 and 2 films, respectively. MF scores tied, so these differences do not establish genuine personalization.

5.5. Controlled Restart Experiments

The baseline fixes user 1 and α = 0.85 (mean Top 10 popularity 245.5). Changing only the restart user to user 2 at α = 0.85 gives mean popularity 254.9 and no common films. Holding user 1 fixed and changing only α to 0.50 gives mean popularity 228.4 and retains seven of ten films. That alpha change replaces Scream, Air Force One and Titanic with Trainspotting, Sense and Sensibility and E.T. The restart vector identifies the user node; that node need not have the highest stationary score. These descriptive results do not establish that a different list has higher relevance, novelty or viewing value.

5.6. Mandatory Question: Keep MF, Replace It with PPR, or Use Both?

I recommend testing PPR as the main "Recommended for you" recommender rather than deploying the degenerate supplied HW6 MF reference. On the identical 861-user test cohort, PPR reached 11.22% Recall@10 versus MF's 2.11%. Users 1 and 2 received contrasting examples: PPR proposed Fargo (Crime/Drama) and Contact (Drama/Sci Fi) for user 1, and Return of the Jedi (Action/Adventure) and Dead Man Walking (Drama) for user 2. The browser MF lists began with Copycat and GoldenEye respectively, but their scores were all tied and therefore did not establish useful preference differentiation. User 1's MF still found 7/19 relevant titles versus PPR's 3/19, showing that the aggregate PPR advantage is not universal. Lower MF list popularity is not evidence of better discovery. The missing original HW5 checkpoint also prevents a conclusive comparison against the previous MF model.

Before rollout, I propose one user randomized A/B test: control receives the current MF main list, treatment receives PPR as the main list. The primary outcome is seven day movie starts per exposed user; guardrails are early abandonment, hides, and title exposure concentration. Fixed user assignment across visits avoids contamination. A functioning production MF comparator is needed before causal conclusions. Offline ratings cannot establish session intent, completed viewing, retention or revenue.

5.7. Limitations and Next Improvement

However, the supplied HW6 MF is a degenerate baseline: every candidate scored 3.580603; validation RMSE was 1.182994 at the selected epoch 1. This is not the original Week 5 selected model, whose checkpoint was unavailable. A separate corrected Week 5 reproduction exists but is not used here. At α = 0.95, 27 validation walks did not converge. User chronological splitting is not globally chronological, and MovieLens ratings are not sessions. Next steps are to recover the previous MF checkpoint, retest several seeds and measure online outcomes.

6. AI Usage Disclosure

AI tools used. OpenCode 1.18.31 assisted coding, browser tests, debugging and evidence packaging. TensorFlow.js is not an AI tool. The supplied original export is seikh_abaidullah_a06_session.json (session ses_ea9d9d477ffeIZmtQoIhprL2sq) [6].

What was checked. OpenCode browser logs document the reference graph, alpha sweep, recommendations and MF checkpoint. Root results.csv and recommendations.csv match the four test Recall@10 values and fixed user lists. Node.js syntax checks passed, but I did not rerun the full browser evaluation, MF training or an online trial. A separate Week 5 reproduction is excluded from the primary table.

Table 7. Evidence Behind the Main Claims

Claim	Evidence source and check
Graph and split	evidence/setup.json; original MovieLens u.data and u.item
Reference fixture and sanity checks	verification.json: 12 / 0 / 0; separate OpenCode log: 33 / 0
Alpha selection and 27 nonconverged walks	evidence/selection.json; validation-095.json
Shared four method test comparison	Root results.csv: all four use 861 test users; same candidate rules
Two user rankings and probabilities	Root recommendations.csv: ID, title, genre, score and popularity
Supplied HW6 MF checkpoint	Saved MF checkpoint: epoch 1; candidate score ties at 3.580603
Reproducibility and AI history	OpenCode integration log (33 / 0); export separate

References

[1] F. M. Harper and J. A. Konstan, “The MovieLens Datasets: History and Context,” ACM Transactions on Interactive Intelligent Systems, vol. 5, no. 4, Article 19, Dec. 2015. doi: 10.1145/2827872. https://doi.org/10.1145/2827872.

[2] S. Brin and L. Page, “The Anatomy of a Large-Scale Hypertextual Web Search Engine,” Computer Networks and ISDN Systems, vol. 30, nos. 1–7, pp. 107–117, 1998. https://doi.org/10.1016/S0169-7552(98)00110-X.

[3] T. H. Haveliwala, “Topic-sensitive PageRank,” in Proceedings of the 11th International Conference on World Wide Web, 2002, pp. 517–526. https://doi.org/10.1145/511446.511513.

[4] Y. Koren, R. Bell, and C. Volinsky, “Matrix Factorization Techniques for Recommender Systems,” Computer, vol. 42, no. 8, pp. 30–37, Aug. 2009. https://doi.org/10.1109/MC.2009.263.

[5] D. Smilkov et al., “TensorFlow.js: Machine Learning for the Web and Beyond,” in Proceedings of SysML, Palo Alto, CA, USA, 2019. https://research.google/pubs/tensorflowjs-machine-learning-for-the-web-and-beyond/.

[6] OpenCode, “Intro: AI Coding Agent Built for the Terminal,” official documentation. https://docs.opencode.ai/docs/. Accessed 29 October 2026.

[7] S. Jin, “Graph-Based Recommendations with Personalized PageRank,” LLM4Rec, HSE University, Week 6 assignment, GitHub repository, 2026. Available: https://github.com/dryjins/RecSys-LLMs/tree/main/week6. Accessed 29 October 2026.

Evidence: HW6_final_report_evidence.zip contains application code, original datasets, test records and root results.csv and recommendations.csv, which match the browser MF and all four Table 3 methods.